Chapter 1: Numerical Methods

We introduce a locally mass conserving numerical framework in this chapter. A conforming mixed finite element discretization (MFEM) is proposed to solve the static Darcy-Stokes coupled methcnics system. The normal flux are captured at the edges of element by the MFEM scheme. The coupled weak form is also adjusted to be locally mass conservative. We solve the transport system with finite volume method (FVM), which is naturally a locally mass conservative scheme.

In this chapter, we first introduction the mesh used in our computation. We present two stable pair for Darcy and Stokes individually equipped with the direct serendipity space. We then apply the defined discrete space to discretize the coupled system. After that, we discuss a noval nonlinear WENO weighting scheme for FVM. Error estimate will be given to MFEM and FVM scheme. We provide a detailed coupling strategy combining MFEM, FVM and eutectic phase package in the end of this chapter.

1.1 Finite Volume Method

The finite volume method (FVM) is a discretization method widely used in various scenarios that conservation laws are involved. The finite volume is a robust algorithm that can be applied to arbitriary complex geometry. The most important aspect of FVM that benefits this simulation is that the local mass conservation is achieved naturally. To begin with, we consider a general conservation law retarding a scalar variable u as

$$\begin{cases} \dfrac{\partial u(\mathbf{x},t)}{\partial t} + \nabla \cdot \mathbf{f}(u;\mathbf{x},t) = 0, & t > 0, \mathbf{x} \in \Omega, \\ u(\mathbf{x},0) = u_0(\mathbf{x}), & \mathbf{x} \in \Omega, \end{cases} \tag{1.1}$$

where we assume the domain $\Omega \subset \mathbb{R}^2$. The term $\mathbf{f}$ is a flux transporting scalar variable u. We write the basic semi-discretized form of basic finite volume method as

$$\frac{\partial \overline{u}}{\partial t} + \frac{1}{|E|} \int_{\partial E} \mathbf{F}(\mathbf{x}, t) \cdot \boldsymbol{\nu} d\mathbf{x} = 0, \tag{1.2}$$

where we use the cell-averaged value $\overline{u} = \frac{1}{|E|} \int_E u(\mathbf{x}, t) d\mathbf{x}$, and $F(\mathbf{x}, t)$ is the approximated numerical flux. We use the Lax-Friedrichs numerical flux

$$\mathbf{F}(u^+, u^-) = \frac{1}{2}[(\mathbf{f}(u^+) + \mathbf{f}(u^-)) \cdot \boldsymbol{\nu} - \alpha_{LF}(u^+ - u^-)], \tag{1.3}$$

where u^+ and u^- represent values approximated from element $E^+ \in \mathcal{T}_h$ and $E^- \in \mathcal{T}_h$ respectively, and $\boldsymbol{\nu}$ is the unit vector pointing from E^+ to E^- orthogonal to the edge. Here, α_{LF} is the Lax-Fredrich stabilizer factor. We have two different ways to compute α_{LF} depending on the knowledge of solution.

- The global LF Stabilizer: $\alpha_{LF} = \max_u |\partial(\mathbf{f} \cdot \boldsymbol{\nu})/\partial u|$;
- The local LF Stabilizer: $\alpha_{LF} = \max(|\partial(\mathbf{f}^+ \cdot \boldsymbol{\nu})/\partial u^+|, |\partial(\mathbf{f}^- \cdot \boldsymbol{\nu})/\partial u^-|)$.

The numerical flux $\mathbf{F}$ (1.3) is consistent and conservtive across the edge. We now replace the u^+ and u^- with reconstructed values and rewrite the numerical flux as

$$\mathbf{F}(\mathcal{R}^+, \mathcal{R}-) = \frac{1}{2}[(\mathbf{f}(\mathcal{R}^+) + \mathbf{f}(\mathcal{R}^-) \cdot \boldsymbol{\nu} - \alpha_{LF}(\mathcal{R}^+ - \mathcal{R}^-)], \tag{1.4}$$

where $\mathcal{R}$ is a higher order reconstruction value. We evaluate the integrals over each edge $e_i \in \partial E$ with a quadrature rule. We represent length of edge e_i and area of cell E with $|e_i|$ and $|E|$ respectively. Assuming a quadrature rule with G points on each edge, we have

$$\overline{u}_t + \sum_{i=0}^{4} \frac{|e_i|}{|E|} \sum_g \omega_{i,g} \mathbf{F}(\mathcal{R}^+_{i,g}, \mathcal{R}^-_{i,g}) = 0, \tag{1.5}$$

where $\omega_{i,g}$ represents gth quadrature points on edge e_i.

1.2 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop a accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for reconstruction of discontinuous values with cell averaged values in **?**. The ML-WENO reconstruction will be primarily used in reconstruction of values on both sided of the edge for numerical flux computation appears in FVM discretization. It can also be applied to general reconstruction of discontinuous variables, e.g., discontinuos darcy pressure appears on the zero-nonzero porosity interface.

1.2.1 Stencil Polynomial

Given a stencil $\mathcal{S} = \{E_0, E_2, \dots, E_k\}$ of the quasiuniform logically rectangular mesh. We can define an associated tensor product stencil polynomial $Q_{s,t} \in \mathbb{Q}_{(s,t)} := \mathbb{P}_{s-1}(x) \otimes \mathbb{P}_{t-1}(y)$, where s and t equal to the stencil sizes in the x- and y-directions, respectively. We scale our stencil polynomial individually as

$$Q_{s,t} = \sum_{p<s,q<t} c_{p,q} \Big(\frac{x - x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p \Big(\frac{y - y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{1.6}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is the center point of the stencil $\Omega_{\mathcal{S}} = \cup_{E\in\mathcal{S}} \subset \mathbb{R}^2$, and $h_{\mathcal{S}} = \sqrt{\sum_k E_k/k}$ is the diameter of average cell size. Let $\xi = (x - x_{\mathcal{S}})/h_{\mathcal{S}}$ and $\eta = (y - y_{\mathcal{S}})/h_{\mathcal{S}}$ be the normalized varaibles, we can rewrite it into normalized version:

$$\hat{Q}_{s,t} = \sum_{p<s,q<t} c_{p,q} \xi^p \eta^q. \tag{1.7}$$

In finite volume scenerio, we are dealing with cell averaged value of the scalar transport variable

$$\overline{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}) d\mathbf{x} = \frac{1}{|E|} \int_E u(x, y) dx dy. \tag{1.8}$$

We can calculate coefficents $c_{p,q}$ by imposing a constraint that cell averaged value (1.8) is preserved through integral with stencil polynomial

$$\frac{1}{|E_k|} \int_{E_k} Q(\mathbf{x}) d\mathbf{x} = \overline{u}_{E_k}, \quad \forall E_k \in \mathcal{S}. \tag{1.9}$$

We will replace $\overline{u}_{E_k}$ with $\overline{u}_k$ in the following discussion for simplicity.

We can represent stencil polynomial with basis polynomials $Q_{s,t} = \sum_k \overline{u}_k \phi_{s,t}^k$, where basis polynomials can be calculated as

$$\frac{1}{|E_{k'}|}\int_{E_{k'}} \phi_{s,t}^k(\mathbf{x})d\mathbf{x} = \frac{1}{|\hat{E}_{k'}|}\iint_{\hat{E}_{k'}} \hat{\phi}_{s,t}^k(\xi,\eta)d\xi d\eta = \begin{cases} 1 & k' = k \\ 0 & k' \neq k \end{cases} \quad \forall (E_{k'}, E_k) \in \mathcal{S}, \tag{1.10}$$

which would lead to a $(s \times t) \times (s \times t)$ non-singular linear system. We can now represent stencil polynomial with normalized basis polynomials as:

$$\hat{Q}_{s,t}(\xi,\eta) = \sum_k \overline{u}_k \hat{\phi}_{s,t}^k(\xi,\eta). \tag{1.11}$$

We define a linear projection operator $\pi_{\mathcal{S}}$ and let $\pi_{\mathcal{S}} u_{\mathcal{S}} = Q$ be a linear projection onto polynomial space W_p such that $\mathbb{Q}_{s,t} \subset W_p$. We estimate interpolation error with the Bramble-Hilbert Lemma **?** and **?** in $H_r(\Omega)$ space. For simplicity, we assume $r = s = t$.

Lemma 1.1. *Let Ω_S be the domain covered by stencil $\mathcal{S}$ in $\mathbb{R}^2$ and let $u \in H^r(\Omega_{\mathcal{S}})$. We have:*

$$\|u - \pi_S u\|^2 \leq h^{2(r-1)} |u|^2_{H^r(\Omega_S)}. \tag{1.12}$$

Proof.

$$\|u - \pi_S u\| = \tag{1.13}$$

$\square$

In a general 2D situtation when $s \neq t$, the execessive order will be "wasted", since the order or accuracy will be $\min(s,t)$. However, in some cases, where we aquire the prior knowledge of the solution being qusi-1D, e.g., the transport velocity is fixed in one direction. We can have uneven stencils that has $\max(s,t)$ order of accuracy but in the desired direction. In other cases when derivative is involved, we also need an stretched stencil to compensate for the order loss when approximates derivative.

1.2.2 Stencil polynomial smoothness indicators

We define a smoothness indicator σ as a measurement of smoothness of $u(\mathbf{x})$ on the stencil S. The classic method from **?** is defined by Sobolev seminorm as:

$$\sigma_P = \sum_{1\le|\alpha|\le m} \int\int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x,y))^2 dxdy, \tag{1.14}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a multi-index, $|\alpha| = \alpha_1 + \alpha_2$, and $\mathcal{D}$ is the derivative operator $\mathcal{D}^\alpha = \partial^{|\alpha|}/\partial x^{\alpha_1}\partial y^{\alpha_2}$. The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$\begin{aligned}
\sigma_P &= \sum_{1\le|\alpha|\le m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x,y))^2 dxdy \\
&= \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_0} \big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi,\eta)\big)^2 d\xi d\eta \\
&= \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'.j'} \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi,\eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi,\eta) d\xi d\eta \\
&= \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'j'} \sigma_{ij}^{i'j'},
\end{aligned} \tag{1.15}$$

where $\sigma_{ij}^{i'j'}$ can be precomputed after mesh is established after mesh is established.

We proposed an alternative way of defining smoothness indicator in **?**.

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{1.16}$$

where $\overline{u}_0$ is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the compuational cost and to deal with distorted stencil on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

We observe that the smoothness indicator exhibit following convergence property that there exists some value $C > 0$ such that when $h \to 0^+$,

$$\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(1) & \text{Not Smooth,} \end{cases} \tag{1.17}$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

1.2.3 ML-WENO weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\{\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \ldots, L, L \geq 0\}$. The reconstruction of u on target cell $E_{i,j}$ is a convex combination of stencil polynomials (1.11)

$$\mathcal{R}(\mathbf{x}) = \sum_l \tilde{\omega}_l Q_l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{1.18}$$

where $\sum_l \tilde{\omega}_l = 1$ is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily chosing a set of linear weights $\{\omega_l\}$ and scale with stencil

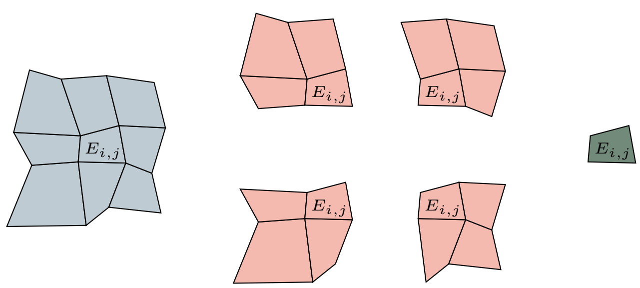

Figure 1.1: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{1.19}$$

where $r_l = \max\{s, t\}$, $\epsilon = 1e-4$ and $a > 0.5$ are picked as constants. The biasing factor $n_l = n(r_l)$ is used to biasing away further from constant level, whose smoothness indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{1.20}$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{1.21}$$

1.2.4 The ML-WENO Reconstruction Error

In this section, we consider the accuracy of our multi-level weighting scheme. We group the set of indices $l \in \mathcal{L} = \{0, 1, 2, \ldots L\}$ according to the order of reconstruction polynomials, i.e., the shape of individual stencil. We denote the set of indices of stencils have s cells in x direction and t cells in y direction as $\mathcal{L}_{s,t}$. We make the assumption that the mesh resolution is fine enough so that there exist at least one stencil that is smooth if shock presents. We define two sets holding indices of stencils with or without jump as

$$\begin{aligned} \text{Smooth} &= \{l : \quad u \text{ is smooth on } \mathcal{S}_l, \} \\ \text{Jump} &= \{l : \quad u \text{ has a jump on } \mathcal{S}_l\} \end{aligned} \tag{1.22}$$

where we assume Smooth $\cup$ Jump $= \mathcal{L}$ and Smooth $\cap$ Jump $= \emptyset$. We represent the set of indices of stencils of shape (s, t) that is smooth as $\mathcal{L}_{s,t} \cap$ Smooth, and the set of stencils of shat (s, t) that has a jump as $\mathcal{L}_{s,t} \cap$ Jump. For convience, we assume square stencils, where $r = s = t$. The accuracy result can be applied to $s \neq t$ situation directly in the selected direction. We show that the reconstruction is accurate with respect to the target cell E_0 to order

$$r_{\max} = \max\{r : \text{Smooth} \cap \mathcal{L}_{r,r} \neq \emptyset\}. \tag{1.23}$$

In Figure 1.2, we illustrate a (3,2,1) combination of stencils with thow shock presents. Here, Smooth $= \{0, 1\}$, which indicates that stencil $\mathcal{S}_0$ and $\mathcal{S}_1$ do not have jump

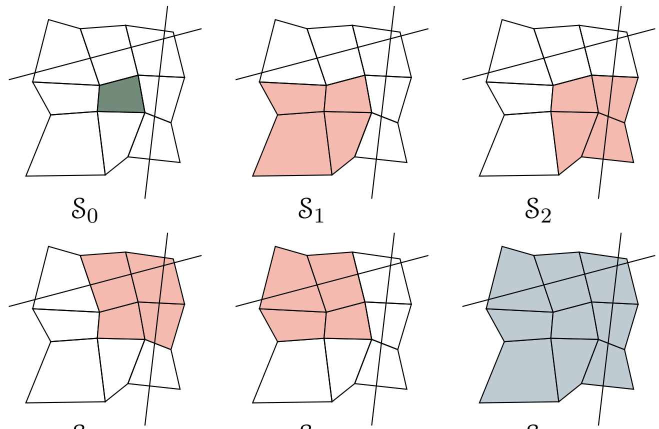

Figure 1.2: An illustration of a (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh with two shock presents.

present. In this case, we expect the accuracy of reconstruction is $r_{\max} = \text{Smooth} \cap \mathcal{L}_{2,2} = 2$.

We state the estimation of reconstruction error show in **?**. We start by estimating the error of our nonlinear scaling (1.19) asscoiated to some stencil $\mathcal{S}_l$ for some $m > 0$.

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m}. \tag{1.24}$$

We estimate the reconstruction error for two situtaions.

- $l \in \text{Smooth}$, the smoothness indicator $\sigma_l = Ch^2 + \mathcal{O}(h^3)$;
- $l \in \text{Jump}$, the smoothness indicator $\sigma_l = \Theta(1)$.

Use the asymptotic estimation result from **?**. Let $n \geq m$, $a > 0$, and $b > 0$

$$\begin{aligned} \frac{a}{bh^m + \mathcal{O}(h^n)} &= \frac{ah^{-m}}{b}\big(1 + \mathcal{O}(h^{n-m})\big), \\ \frac{a}{bh^{-n} + \mathcal{O}(h^{-m})} &= \frac{ah^n}{b}\big(1 + \mathcal{O}(h^{n-m})\big). \end{aligned} \tag{1.25}$$

For the smooth stencil $\mathcal{S}_l \in$ Smooth

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m} = \frac{\omega_l}{((C+\epsilon)h_0^2 + \mathcal{O}(h^3))^m}$$
$$= \omega_l \Big(\frac{h_0^{-2}}{C+\epsilon}(1+\mathcal{O}(h)) \Big)^m \tag{1.26}$$
$$= \frac{\omega_l}{(D+\epsilon)^m} h_0^{-2m}(1+\mathcal{O}(h)).$$

In conclusion, we estimate the nonlinear scaling as

$$\hat{\omega}_l = \begin{cases} \dfrac{\omega_l}{(C+\epsilon)^m} h_0^{-2m}(1+\mathcal{O}(h)), & \text{if} \quad l \in \text{Smooth}, \\ \Theta(1), & \text{if} \quad l \in \text{Jump}. \end{cases} \tag{1.27}$$

Then we estimate the error for the normalized nonlinear weighting (1.21). We expand the sum of nonlinear scaling as

$$\sum_l \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l + \sum_{l \in \text{Jump}} \hat{\omega}_l. \tag{1.28}$$

We estimate the smooth part of the nonlinear weighting as

$$\sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \frac{\omega_l}{(C+\epsilon)^m} h_0^{-2(ar+\eta)}(1+\mathcal{O}(h)). \tag{1.29}$$

Assume $h < 1$, we keep the largest $ar + \eta(r)$ number as our estimator

$$\begin{aligned} \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = & \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))}(1+\mathcal{O}(h)) \\ & \qquad + \Theta(h^{-2(ar+\eta(r))}) \\ = & \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))} \\ & \qquad + \mathcal{O}(h^{1-2(ar_{\max}+\eta(r_{\max}))}). \end{aligned} \tag{1.30}$$

later, we denote the set Smooth $\cap\, \mathcal{L}_r$ as Smooth$_r$ for simplicity. The overall accuracy estimation of the nonlinear weighting (1.21) is presented as

- $l \in$ Smooth

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}} = \frac{\omega_l}{\sum_{l' \in \text{Smooth}_r} \omega_{l'}} h^{2[a(r_{\max}-r)+\eta(r_{\max})-\eta(r)]}(1+\mathcal{O}(h)); \tag{1.31}$$

- $l \in$ Jump

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}} = \Theta(h^{2[a(r_{\max})+\eta(r_{\max})]}). \tag{1.32}$$

Now we can estimate the reconstruction error as

$$\begin{aligned}
|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| &= |\sum_l \tilde{\omega}_l (u(\mathbf{x} - Q_l(\mathbf{x})))| \\
&\le \sum_{l \in \text{Smooth}} \tilde{\omega}_l |u(\mathbf{x}) - Q_l(\mathbf{x})| + \sum_{l \in \text{Jump}} \tilde{\omega}_l |u(\mathbf{x}) - Q_l(\mathbf{x})| \\
&\le \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \Theta(h^{2[a(r_{\max}-r)+(\eta(r_{\max})-\eta(r))]})\mathcal{O}(h^r) \\
&\qquad + \sum_{l \in \text{Jump}} \Theta(h^{2(ar_{\max}+\eta(r_{\max}))})\mathcal{O}(1) \\
&= \mathcal{O}(h^{r_{\max}})
\end{aligned} \tag{1.33}$$

Combining everything above and (1.18), we arrive at the formal error estimate of the reconstruction.

Lemma 1.2. *Let $\mathcal{R}(\mathbf{x})$ be the reconstruction. Assuming a qusi-uniform mesh, $h_0 = \Theta(h)$, there exists a constant $C > 0$ such that for any $\mathbf{x} \in E_0$,*

$$|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| \le Ch^{r_{\max}}, \tag{1.34}$$

where $r_{\max}$ is defined by (1.23).

1.2.5 Handling Boundary Conditions

In our simulation, we couple FEM and FVM solver on a single mesh. Therefore, accurate and compatible boundary condition should be assigned accordingly. However, it is usually difficult to manage boundary condition for higher order finite volume and finite difference method, which require wide stencils. Some deal with the difficulty use ghost cells or points outside the boundary, and maintain the same stencil as if they were in the interior of the domain. However, the extrapolation or estimation of the manufactured values of ghost cells or ghost points are not straightforward. A notable method, the inverse Lax-Wendroff procedure, substitute the normal derivative of the solution with tangential and time derivatives using the PDE relation in an iteratvie way to impose more precise values for the ghost cells or points. The key to the above strategy is to have accurate and stable reconstruction near the boundary. ML-WENO gives the flexibility to add stenncils of the desired size that span back into the domain.

In order to assign proper boundary condition to a FVM discretization, we should determine whether the flow is inflow or outflow first. If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} > 0$, we recognize that it is a outflow boundary, therefore, no fixed boundary condition should be assigned. We use "free outflow" boundary that the numerical flux is simply $\mathbf{f}(u_B) \cdot \boldsymbol{\nu}$.

The inflow boundary can be understood in various ways. We can understand it as prescribing the inflow flux $\mathbf{f}_B$ or fixing the boundary value u_B as an equivelent to Dirichlet boundary in the context of Finite Element. In the special case when a zero flux is prescribed, we would call it noflow boundary. In the inflow case, there may be a discontinuous inflow shock wave. We require a swift response inside the domain to align with the specified Dirichlet value. We therefore add an extra constant stencil polynomial to the inflow boundary to accommodate the potential discontinuity.

There are some other special cases that require special treatment on the boundary, e.g., when symmetry should be enforced.

We now present two examples

1.2.6 Numerical Test

For completeness, we test our ML-WENO algorithm with simple Riemann shock and rarefraction. Let $\mathbf{f}(u) = (u^2/2, u^2/2)$ be the Burgers flux function. In this example, we take the initial condition $u_0(x,y) = 1$, and the computational domain $\Omega = [0,3] \times [0,1]$. The true solution is

$$u(x,y,t) = \begin{cases} 0, & x < 0.5, x \geq 0.5t + 1.5, \\ (x-0.5)/t, & 0.5 \leq x < t + 0.5, \\ 1, & t + 0.5 \leq x < 0.5t + 1.5. \end{cases} \tag{1.35}$$

We forward martching to the time = 2.0, when trailing rarefaction reaches the leading shock. We use SSP2RK time stepping scheme.

1.3 Coupling Computation

Now we couple locally mass conserved MFEM and FVM method together in this section. The remaining uncertainty of porosity ϕ will be fulfilled by phase package evaluation. Before we throw ourselves into the fully coupled problem. We test another porosity evolution scenerio with prescribed melting. Then we will show an example combining MFEM, FVM and eutectic phase package alltogether.

1.3.1 Coupled Algorithm

Let $U^n = (C^n, H^n)$ be the solution to the transport system at time t^n. Let $V^n = (\check{\mathbf{v}}^n_{h,r}, \check{\mathbf{v}}^n_{h,s}, \check{q}^n_{h,l}, \check{q}^n_h)$ denote the solution to the Darcy-Stokes system at the same time. Moreover, let $W^n = (\phi^n, \check{T}^n, c^n_s, c^n_l, \phi^n_1)$ denote the output of the phase calculation at t^n. We formulate an overall solution methods. The coupled solver is initialized from the initial condition on the transport variables U^0 specifying the total initial mass and energy. The eutectic phase package gives W^0. Once we know W^0, we have phase related variables porosity and individual concentrations.

1.3.2 Prescribed Melting Problem

In this section, we illustrate a simpler model coupling case coupling MFEM and FVM solver. We prescribe the melting rate or the change of porosity as a fixed

Algorithm 1 Coupled Solver

Let U^0 be given as initial data.

1: **for** Time level $n = 0, 1, \ldots, n_{\max} - 1$ **do**

2: Given U^n, compute W^n by the eutectic phase package.

3: Given ϕ_f^0 in W^0, solve the Darcy-Stokes system (**??**)–(**??**) for V^0.

4: Given W^n and V^n, compute effective velocity (**??**) and phase averaged velocity as

$$\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i}, \quad \overline{\mathbf{v}} = \phi \mathbf{v}_r + \mathbf{v}_s.$$

and then solve the transport system (**??**) for U^{n+1}.

5: **end for**

function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{1.36}$$

In this case, we transport porosity directly considering full advection as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot \overline{\mathbf{v}} = 0. \tag{1.37}$$

1.3.3 Fully coupled Computation

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